Abdelrahman Khalifa

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Education

Minia University — Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering — *2020 – 2025*

- Grade: Very Good with Honors. Graduation project: AUTOSAR V2X vehicle on STM32 with a dual-bank bootloader.
- Activities: Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.

Experience

Google Summer of Code Contributor — May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) — *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app: printers grouped by device, supply levels, print queues, live updates and an Add Printer flow for legacy printers via PAPPL (cosmic-settings #2182, in review).
- Designed the stack as 5 Rust crates: `printers-core` (types, grouping queues into physical devices), `printers-client` (async varlink client), `printers-server` (CUPS/IPP), `printers-ui` (iced) and `printers-app`.
- Ran the backend as a varlink IPC service in `cosmic-settings-daemon` (cosmic-settings-daemon #188, in review), so the C FFI code never loads in the Settings process; the standalone app falls back to running it in-process.

Embedded Software Engineer — Aug 2025 – Present
EKSON — *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Developed the PyQt5 desktop control app, cutting CPU usage and latency by about 40% (measured) using asyncio to wait for ESP32 and server messages concurrently; added remote control and firmware updates through a Node.js/MongoDB backend.

Embedded Linux Systems Intern — Dec 2024 – Jan 2025
Orange Digital Center (ODC) — *Egypt*

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

Projects

OpenPrinting cups-rs (Rust) and libcups (C) | *Upstream contributions (all accepted)* — 2026

- cups-rs: ported to libcups3 (CUPS 2 and 3 builds, CI); added IPP management operations, Printer Application support and FFI bindings for the libcups3 DNS-SD API (PRs #18–#22).
- libcups: fixed crashes in `cupsDNSSDDelete()` and a leaked client in the Avahi DNS-SD code (#164, #165).

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* — Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).

Kernel Monitor | *C, Linux, ARM, QEMU* — GitHub

- Character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time to user space; run on QEMU.

IoT Device Communication System | *C++, Yocto* — GitHub

- TCP and UDP multicast between a Pi and QEMU clients.

Technical Skills

Languages: Rust, C, Modern C++, Python (asyncio), Bash, JavaScript (Node.js), Java
Systems & Linux: Linux (Ubuntu/Debian), operating systems, FFI and bindings, IPC (varlink, D-Bus; zbus, zlink), concurrency (async/event-driven, multithreading), memory management, networking (TCP/UDP), CUPS/IPP, DNS-SD/mDNS (Avahi), kernel modules and device drivers, firmware and bootloaders
Tooling: Cargo, Clippy, rustfmt, Git/GitHub, GitHub Actions CI, CMake, Make, Yocto, Buildroot, QEMU, testing, debugging, performance optimization, code review